(3) **tenet + ti = metsy̰ maben pungu**

(4) **metsy̰ maben trok × pungu = lir anasy̰**

(5) **teri ana × ana + roky̰ri asy̰m = telang ty̰ko maben tenet**

(6) **semy̰r maben ti + liri ana = tenem ser metsy̰**

(7) **ty̰ko × trok = tenemri pezy̰**

(8) **lir anasy̰ri pezy̰ - lir maben pungu = tenem maben ty̰ko**

(9) **teri asy̰m × trok = lir anasy̰ maben ti**

(a) Write the equalities (1–9) in digits, given that metsy̰ maben ti is 18.

(b) Write in Chungli Ao: 10, 31, 36, 58, 93. If an exact expression cannot be determined from the given data, write out all possibilities.

⚠ Chungli is one of five dialects of the Ao language, which belongs to the Sino-Tibetan family. It is spoken by approx. 130,000 people in Nagaland, India. y̰ is a vowel.

Problem 64. Word Embeddings (PLO, 2020)

Word embeddings are numeric representations of words as points in d-dimensional space. That is, a word can be represented using d numbers, where each number represents the coordinate of that number along a particular dimension.

For example, 2-dimensional embeddings are representations of words as points on a plane using a pair of numbers. The first number corresponds to the x-coordinate of the point and the second number to the y-coordinate of the point.

These word embeddings are very useful representations of words for training machine learning models to perform natural language processing tasks. The embedding for a word is computed by looking at the word in a large corpus of text, and looking at what words occur in the context of that word.

Some machine translation models use embeddings of words in the source language to compute the embedding of the word in the target language, and then look up the word corresponding to that embedding. In this problem, we consider translating **Hrangkhol** to English.

Here are some English words with their embeddings in the 2-D space:

bull	**[2, 2]**	*son-in-law*	**[-5, 1]**
cow	**[3, 3]**	*daughter-in-law*	**[-4, 2]**
mare	**[4, 5]**	*grandson*	**[-6, 3]**
(female) duck	**[3, -1]**		

Here are some Hrangkhol words with their embeddings and their corresponding English translations in random order:

1.	**vôkchal [-4, 6]**	A.	*male parrot*
2.	**vôkpui [-6, 8]**	B.	*hen*
3.	**sakorchal [-8, 6]**	C.	*sow*
4.	**arkhong [4, 2]**	D.	*daughter*
5.	**arpui [2, 4]**	E.	*boar*
6.	**vatôkhong [4, 4]**	F.	*granddaughter*
7.	**vaikikhong [8, 4]**	G.	*drake*
8.	**naipa [-4, -12]**	H.	*stallion*
9.	**nainu [-6, -10]**	I.	*son*
10.	**tunu [-8, -10]**	J.	*rooster*

(a) Match the Hrangkhol words with their English translations

(b) Write the Hrangkhol word embeddings for the English words given with their embeddings

(c) Write the word embeddings of the English translations *A-J*

(d) Knowing that **kêl** = *goat* and **tar** = *old person* translate the following words to Hrangkhol:

11. *female parrot*
12. *he-goat*
13. *old woman*

Problem 65. Lele (MSK, 2017)

Arithmetic examples are given in the Lele language*. All numbers in the examples are less than 100:

(1) **manonoh × manondoruoh = mahangul e ondoruoh**
(2) **manondrsih × manondrsih = manondorungul e sih**
(3) **manonoh × masungul = manonngul**
(4) **matoloh + marngul e ruoh = marngul e limah**
(5) **malimngul e onoh + masungul e hahou = manondrtilngul**

(a) Write in Lele:

2, 7, 65, 99

(b) Write in numbers:

(6) **matilngul e toloh + mahahou = matilngul e ondrtoloh**